Preface

Albert Einstein, who was in many ways the father of quantum mechanics, had a notorious love-hate relation with the subject. His debates with Niels Bohr—Bohr completely accepting of quantum mechanics and Einstein deeply skeptical—are famous in the history of science. It was generally accepted by most physicists that Bohr won and Einstein lost. My own feeling, I think shared by a growing number of physicists, is that this attitude does not do justice to Einstein's views.

Both Bohr and Einstein were subtle men. Einstein tried very hard to show that quantum mechanics was inconsistent; Bohr, however, was always able to counter his arguments. But in his final attack Einstein pointed to something so deep, so counterintuitive, so troubling, and yet so exciting, that at the beginning of the twenty-first century it has returned to fascinate theoretical physicists. Bohr's only answer to Einstein's last great discovery —the discovery of *entanglement*—was to ignore it.

The phenomenon of entanglement is the essential fact of quantum mechanics, the fact that makes it so different from classical physics. It brings into question our entire understanding about what is *rea*l in the physical world. Our ordinary intuition about physical systems is that if we know everything about a system, that is, everything that can in principle be known, then we know everything about its parts. If we have complete knowledge of the condition of an automobile, then we know everything about its wheels, its engine, its transmission, right down to the screws that hold the upholstery in place. It would not make sense for a mechanic to say, "I know everything about your car but unfortunately I can't tell you anything about any of its parts."